Computer Maintenance Basics

Computer Longevity/Durability

Computer Longevity

General Care

- Keep food or liquids away from your laptop/computer to prevent unnecessary damage.
- Plug into a surge protector
- When not in use turn off or sleep your computer.

Computer Longevity (Laptop)

<u>Battery Care</u>

- Don’t keep a laptop plugged in all the time
- Don’t expose to extreme temperatures
- Avoid fully draining the battery.

Computer Longevity (Laptop)

Temperature

- Operate your computer in a cool, dry and well ventilated area
- Cooling pads will extend the life of your laptop
- Do not place a laptop on…
  - Bed
  - Blanket
  - Unstable furniture
  - Near the edge of anything
  - Anything that will block the fans

Computer Longevity (Desktop)

<u>Desktop placement</u>

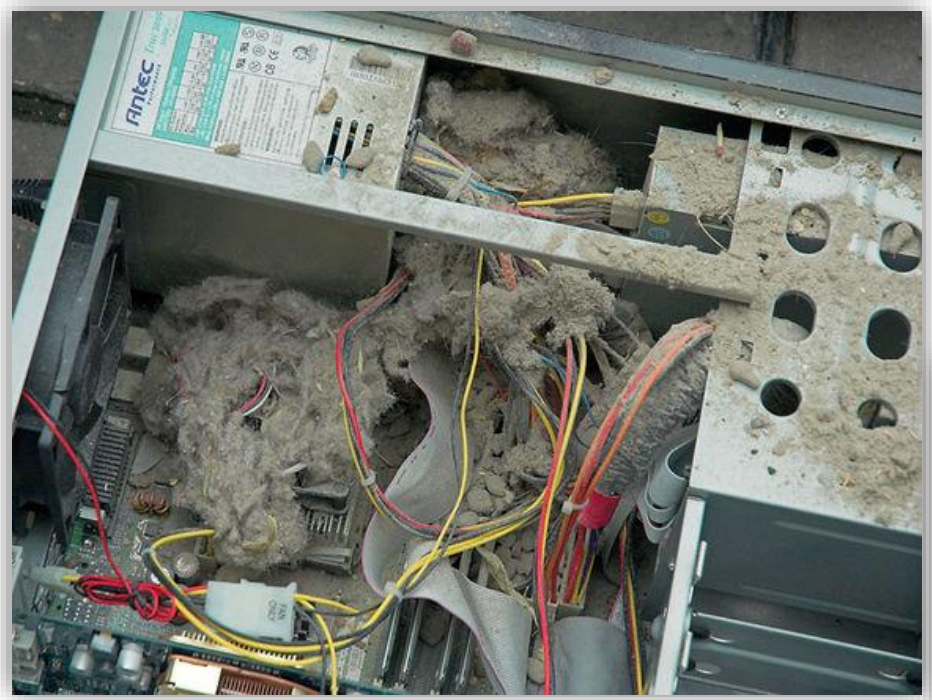
- Operate your computer in a well ventilated, cool and dry area
- Do not cover up fans with books, papers, etc.
- Do not place where it can easily fall or be hit

<u>Locations to avoid</u>

- Under your desk
- Near a window
- Near a heat vent

Hardware Maintenance

Hardware Maintenance



Don’t let your computer look like this!

Hardware Maintenance

- Warning! Before doing any cleaning or work turn off and unplug your computer.

- Use an air compressor or compressed air canisters to remove dust.
  - Remove the side panel and blow dust off components.

Hardware Maintenance

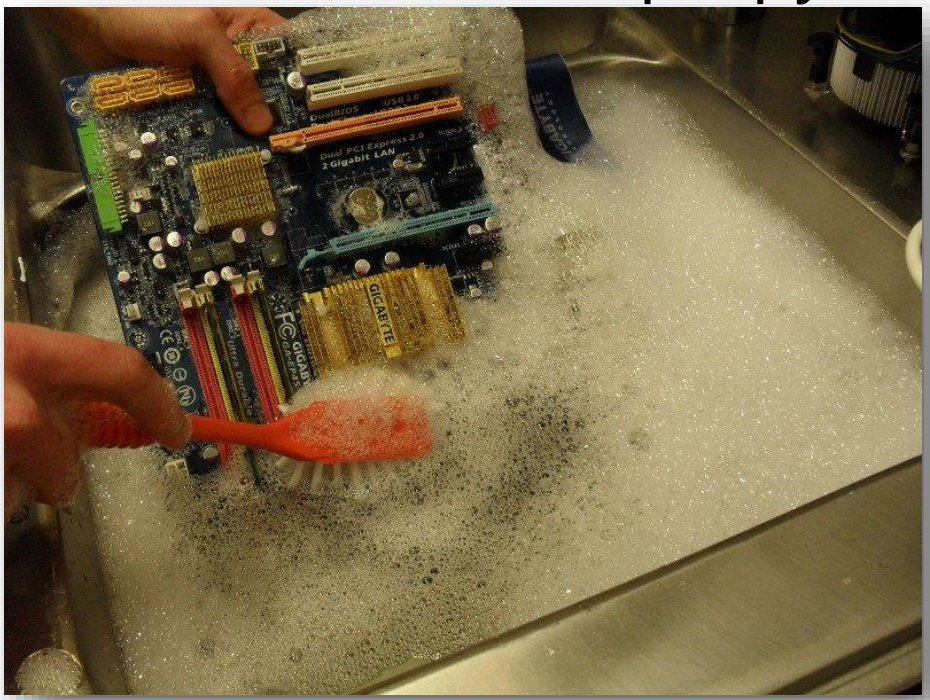
<u>Screen Cleaning</u>

- Clean your screen with a microfiber cloth and screen cleaning solution. Apply solution to your cloth and gently wipe.

- Use electronics cleaning solution or make your own
  - You can make your own solution with one part 70% isopropyl alcohol and one part water.

Hardware Maintenance

- In extreme cases, components can be washed with soap and water, a gentle dish washer, or isopropyl alcohol 70% or greater.
- Do not wash!
  - Laptops
  - Monitors
  - Power supplies
  - Hard drives
  - Batteries



- Let your components dry **100%** before using them to avoid damage.

Laptop Protection

Laptop Protection

Traveling with a laptop

- Purchase a laptop bag or cover
- Properly pack the laptop safely before travel
- Keep the laptop secure while traveling
- Keep in the climate controlled cabin

Software

Software

Windows XP and Vista are no longer supported with any security updates!

- If possible purchase a new computer or upgrade to a newer version of Windows
- If not possible only use the computer offline, or be very careful about safe internet usage

Internet Browsers

Best Internet browsers for security and speed

- Firefox, Chrome, Opera, or Edge

Updates

- Set all of your updates to automatic to avoid forgetting to update them.
  - Windows
  - Web browsers
  - Antivirus software
  - Java

Viruses

What is a Virus?

A Virus is a malicious set of code meant to harm you and/or your computer

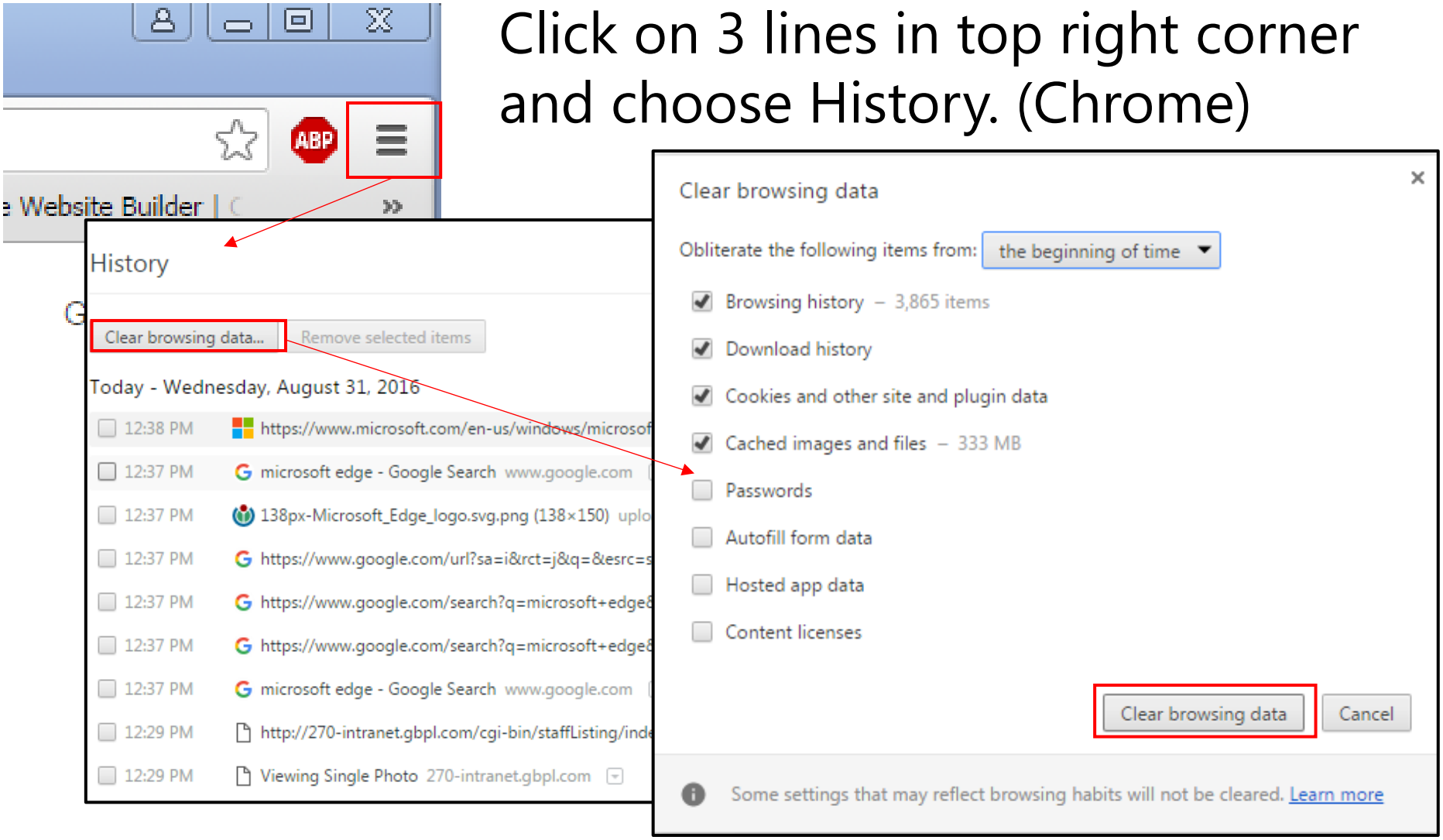
<u>Preventing viruses</u>

- Windows Defender
  - Comes with free with Windows 8 and higher
- Paid antiviruses
  - Kaspersky
  - BitDefender Plus
  - Norton
  - McAfee

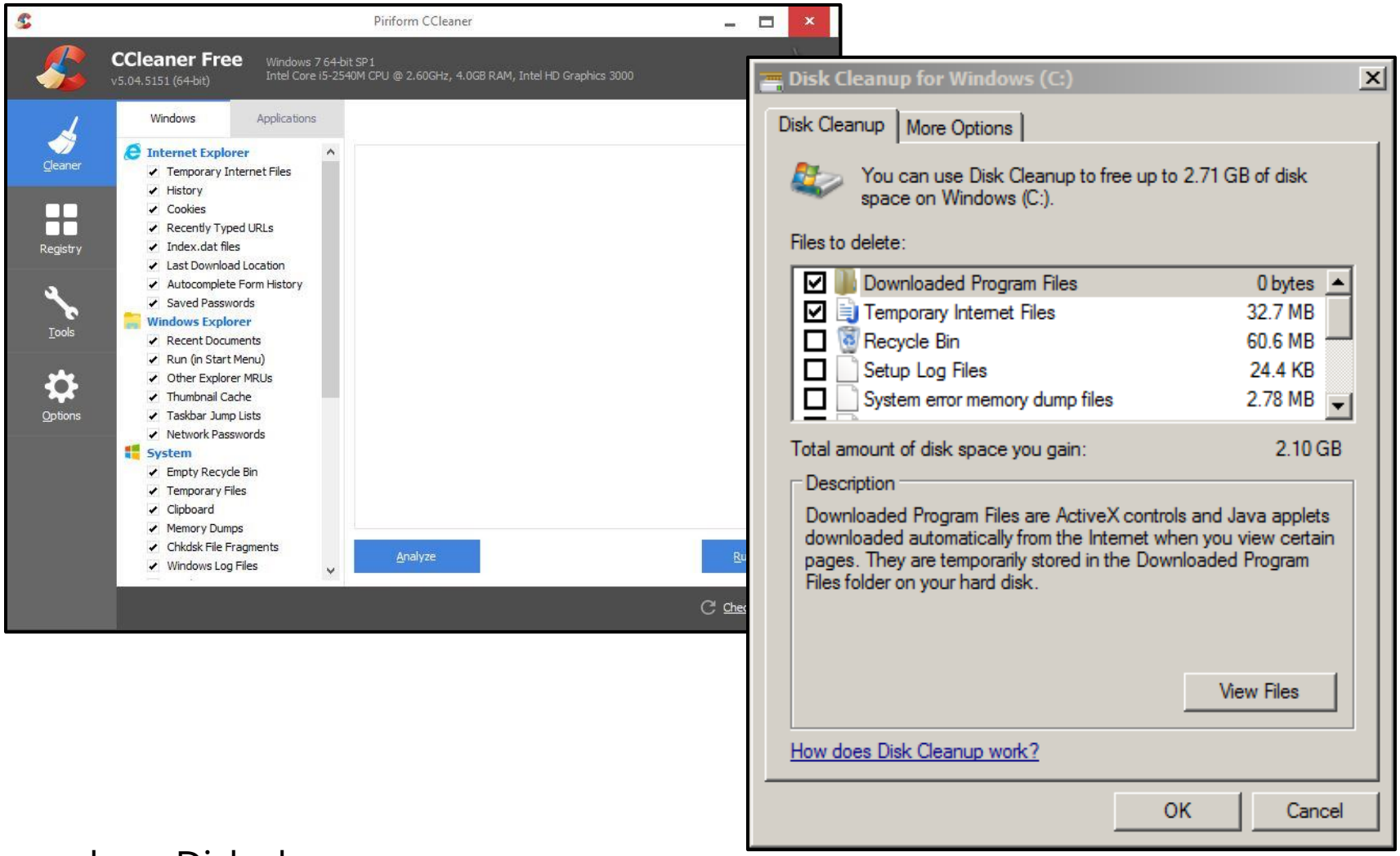
Performance

Clearing History - Chrome

Click on 3 lines in top right corner and choose History. (Chrome)

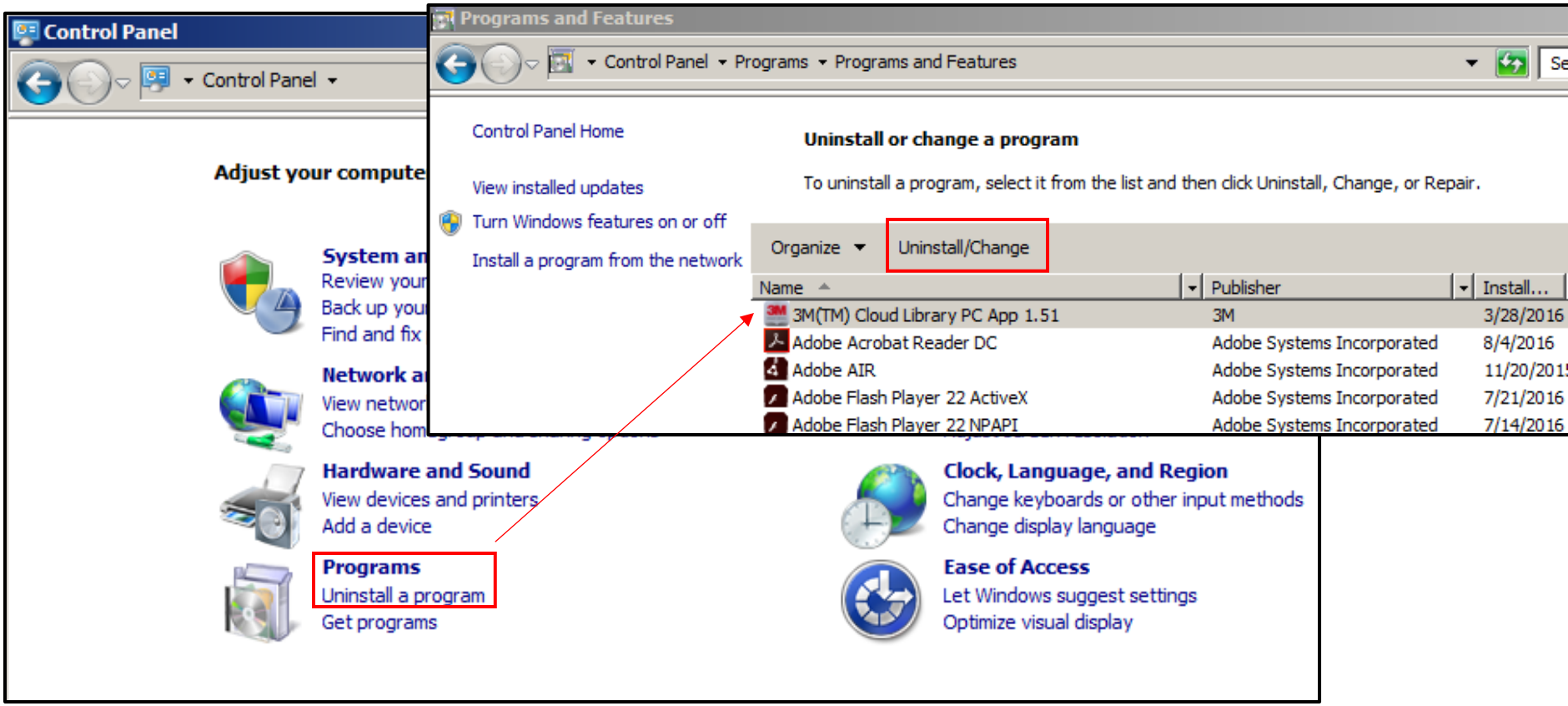


Ccleaner / Disk Cleanup



In search -> Disk cleanup ....

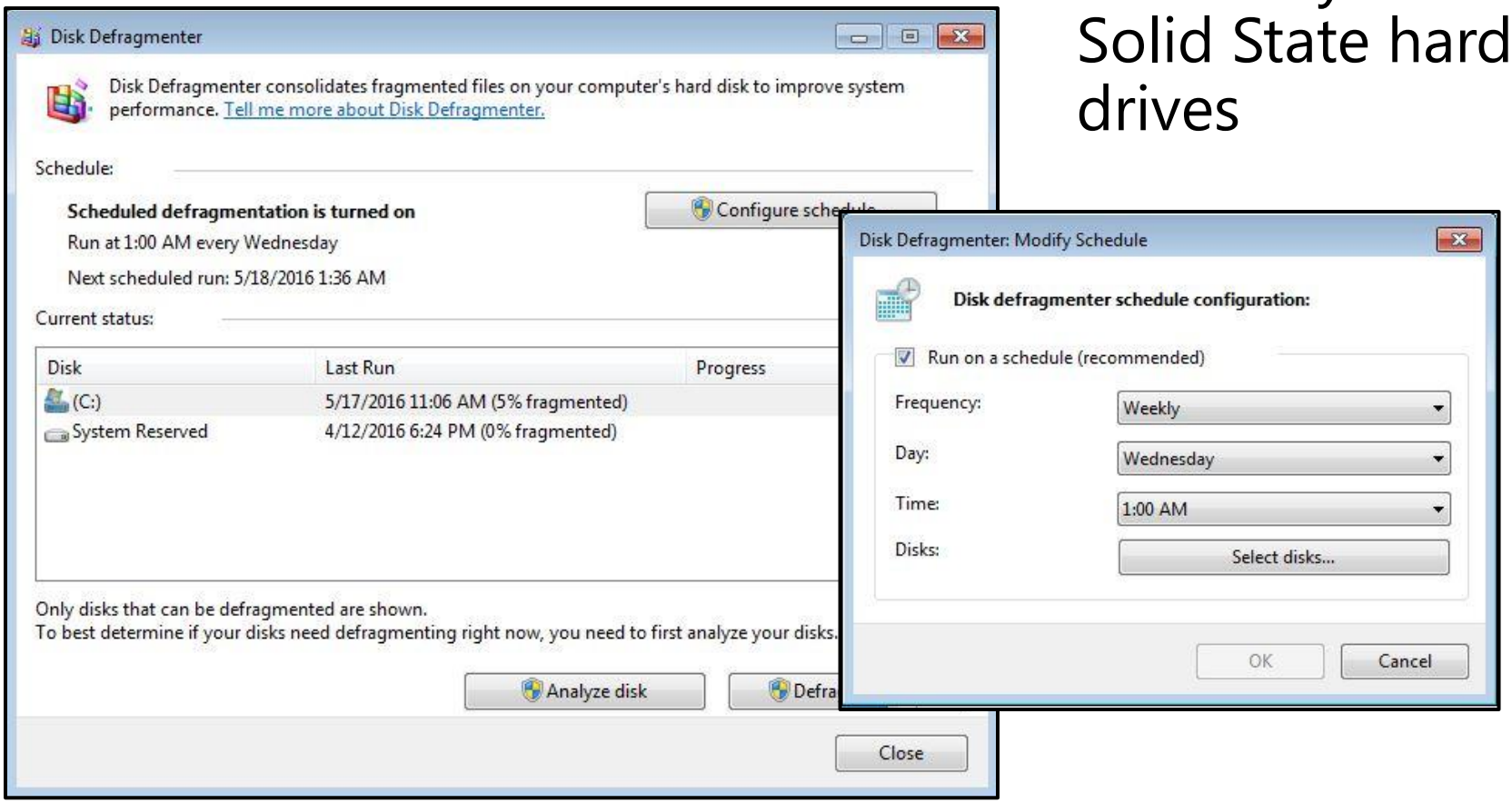
Deleting unwanted programs



- Uninstall programs you never use to free up hard drive space

Defragmentation

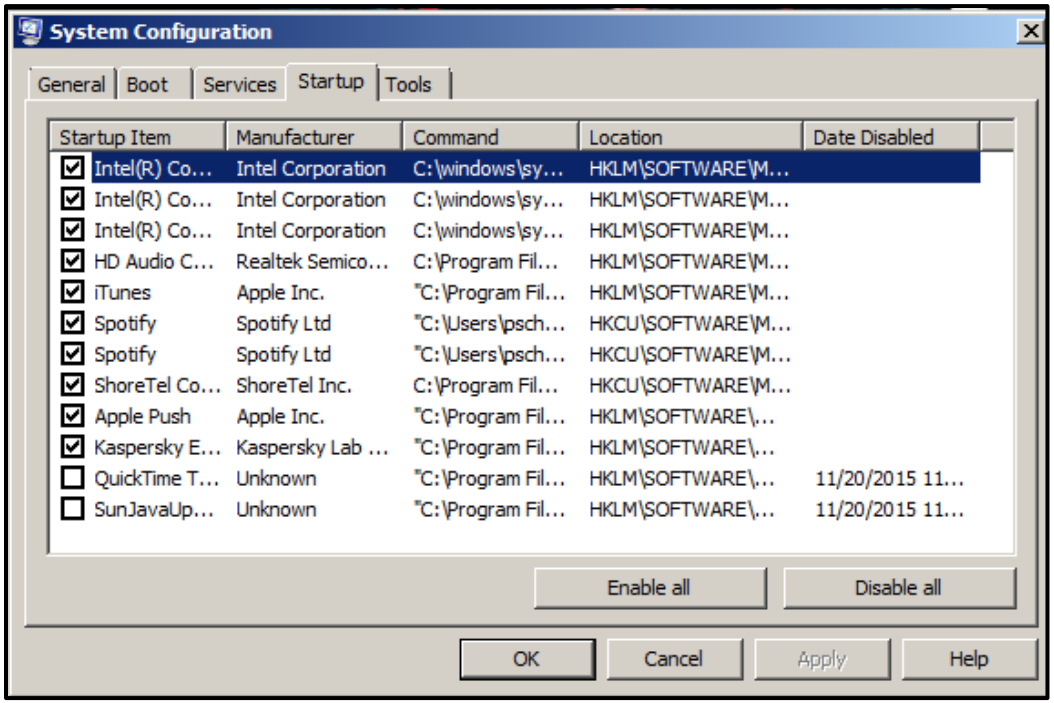
Note: not necessary for Solid State hard drives



Task Manager

How many start up programs are running?

- How to access the startup programs?
  - Windows 7 msconfig
  - Windows 8-10 in the task manager



Backup

Cloud Storage

- iCloud
- Google Drive
- One Drive
- Drop Box
- Amazon cloud drive

iCloud

Google Drive

One Drive

Drop Box

Amazon Cloud Drive

Q.

Which of the following resources is the MOST critical when using cloud file-storage synchronization applications?

A. Disk speed

B. I/O bandwidth

C. RAM utilization

D. CPU utilization

Q.

Which of the following features will save the battery life of a laptop significantly if they are disabled? (Choose two.)

A. Touchpad

B. Volume

C. Screen orientation

D. Cellular

E. Bluetooth

F. Fingerprint reader

Q.

What is the difference between **Backup** vs **Archive**?

Alternative Backup Sources

Physical Back up

- Backup up important files to an external Hard drive or large Flash drive often (manually)
  - Flash drive 32-128gb
  - External hard drive 500gb-4tb

Paid backup

- Carbonite (cloud storage)
- Novastor Backup (cloud and/or physical storage)
- Acronis Backup (cloud and/or physical storage)

Maintenance schedule

System	Maintenance Times	Notice
Network Services	8 am – 8 pm, Sundays, only as needed	Will provide 24-48 hours' notice when taking down our Novell servers.
Servers	5:00 am to 7:30 am, Thursdays *Typically, windows servers will be patched either the third or fourth Thursday of each month.*	When a longer down time is required, will provide 24-48 hours' notice.
Networking and Internet Infrastructure (including wireless system)	5:00 am to 7:30 am, Thursdays	Will provide 24-48 hours' notice when taking down the phone/ voicemail service.
Telephone	No weekly schedule required	Will provide 24-48 hours' notice when taking down the phone/ voicemail service.
VMWARE Infrastructure	No weekly schedule required	When maintenance is required, affected users will be contacted

- End -